

## DBMS – PRODUCT AND CATEGORY MANAGEMENT

### Project Title: Etsy Handmade Marketplace Database Management System

## 1. Database Design

### 1.1 Category Table

| Attribute            | Data Type    | Constraint  |
|----------------------|--------------|-------------|
| <i>Category_ID</i>   | INT          | Primary Key |
| <i>Category_Name</i> | VARCHAR2(30) | Unique Key  |
| <i>Description</i>   | VARCHAR2(60) | -           |

### 1.2 Product Table

| Attribute           | Data Type    | Constraint   |
|---------------------|--------------|--------------|
| <i>Product_ID</i>   | INT          | Primary Key  |
| <i>Product_Name</i> | VARCHAR2(50) | -            |
| <i>Category_ID</i>  | INT          | Foreign Key  |
| <i>Supplier_ID</i>  | INT          | -            |
| <i>Brand</i>        | VARCHAR2(30) | -            |
| <i>Price</i>        | NUMBER(10,2) | Check (> 0)  |
| <i>Stock</i>        | INT          | Check (>= 0) |
| <i>Expiry_Date</i>  | DATE         | -            |

## 2. Primary and Foreign Key Relationship

Category → Product

- Category.Category\_ID → Primary Key
- Product.Product\_ID → Primary Key
- Product.Category\_ID → Foreign Key referencing Category.Category\_ID
- Relationship: One-to-Many (1:M) — one category contains many products

## 3. Table Creation

### 3.1 Create Category Table

```
CREATE TABLE Category (  
    Category_ID INT PRIMARY KEY,  
    Category_Name VARCHAR2(30) UNIQUE,  
    Description VARCHAR2(60)  
);  
  
Table created.
```

The Category table is created with Category\_ID as the primary key and Category\_Name as a unique field.

### 3.2 Create Product Table

```
CREATE TABLE Product (  
    Product_ID INT PRIMARY KEY,  
    Product_Name VARCHAR2(50),  
    Category_ID INT REFERENCES Category(Category_ID),  
    Supplier_ID INT,  
    Brand VARCHAR2(30),  
    Price NUMBER(10,2) CHECK (Price > 0),  
    Stock INT CHECK (Stock >= 0),  
    Expiry_Date DATE  
);  
  
Table created.
```

The Product table is created with Product\_ID as the primary key and Category\_ID as a foreign key referencing the Category table, establishing a one-to-many relationship.

## 4. Data Insertion

### 4.1 Category Records

Ten category records are inserted into the Category table, followed by a commit to save the transaction.

```
INSERT INTO Category VALUES (1, 'Jewelry', 'Handmade jewelry and accessories');
1 row created.

INSERT INTO Category VALUES (2, 'Home Decor', 'Handcrafted home decor items');
1 row created.

INSERT INTO Category VALUES (3, 'Clothing & Accessories', 'Handmade clothing and accessories');
1 row created.

INSERT INTO Category VALUES (4, 'Art & Collectibles', 'Original art and collectible pieces');
1 row created.

INSERT INTO Category VALUES (5, 'Bath & Beauty', 'Natural bath and beauty products');
1 row created.

INSERT INTO Category VALUES (6, 'Wedding & Party Supplies', 'Wedding and party decor supplies');
1 row created.

INSERT INTO Category VALUES (7, 'Toys & Games', 'Handmade toys and games');
1 row created.

INSERT INTO Category VALUES (8, 'Craft Supplies & Tools', 'Craft supplies and tools');
1 row created.

INSERT INTO Category VALUES (9, 'Vintage Finds', 'Vintage and antique items');
1 row created.

INSERT INTO Category VALUES (10, 'Paper & Party Goods', 'Paper goods and party printables');
1 row created.
```

## 4.2 Product Records

One hundred product records are inserted into the Product table, ten per category, each linked via Category\_ID, followed by a commit.

### Category 1 — Jewelry

```
INSERT INTO Product VALUES (101, 'Beaded Necklace', 1, 1001, 'LunaCraftsCo', 850, 25, NULL);
1 row created.

INSERT INTO Product VALUES (102, 'Silver Hoop Earrings', 1, 1002, 'SilverMoonStudio', 650, 30,
NULL);
1 row created.

INSERT INTO Product VALUES (103, 'Charm Bracelet', 1, 1003, 'WhimsyWorks', 720, 20, NULL);
1 row created.

INSERT INTO Product VALUES (104, 'Wire Wrapped Ring', 1, 1004, 'StoneAndWireCo', 480, 35, NULL);
1 row created.

INSERT INTO Product VALUES (105, 'Pearl Pendant', 1, 1005, 'OceanPearlCraft', 950, 15, NULL);
1 row created.

INSERT INTO Product VALUES (106, 'Beaded Anklet', 1, 1006, 'BohoBeadsShop', 380, 40, NULL);
1 row created.

INSERT INTO Product VALUES (107, 'Gemstone Choker', 1, 1007, 'CrystalCoveCraft', 1100, 12,
NULL);
1 row created.

INSERT INTO Product VALUES (108, 'Layered Chain Necklace', 1, 1008, 'GoldenThreadJewels', 890,
18, NULL);
1 row created.

INSERT INTO Product VALUES (109, 'Macrame Bracelet', 1, 1009, 'KnotsAndBeads', 320, 45, NULL);
1 row created.

INSERT INTO Product VALUES (110, 'Statement Earrings', 1, 1010, 'VividVineDesigns', 560, 22,
NULL);
1 row created.
```

## Category 2 — Home Decor

```
INSERT INTO Product VALUES (111, 'Macrame Wall Hanging', 2, 1011, 'KnotsAndBeads', 1450, 12,
NULL);
1 row created.

INSERT INTO Product VALUES (112, 'Ceramic Planter', 2, 1012, 'ClayAndPetalStudio', 780, 20,
NULL);
1 row created.

INSERT INTO Product VALUES (113, 'Wooden Coaster Set', 2, 1013, 'WoodWhisperCraft', 550, 30,
NULL);
1 row created.

INSERT INTO Product VALUES (114, 'Hand-painted Vase', 2, 1014, 'BrushAndClayArt', 1650, 8,
NULL);
1 row created.

INSERT INTO Product VALUES (115, 'Woven Basket', 2, 1015, 'WillowWeaveCo', 920, 15, NULL);
1 row created.

INSERT INTO Product VALUES (116, 'Candle Holder', 2, 1016, 'FlameAndForm', 600, 25, NULL);
1 row created.

INSERT INTO Product VALUES (117, 'Dreamcatcher', 2, 1017, 'FeatherDreamsCraft', 480, 28, NULL);
1 row created.

INSERT INTO Product VALUES (118, 'Wall Tapestry', 2, 1018, 'TextileTalesShop', 1350, 10, NULL);
1 row created.

INSERT INTO Product VALUES (119, 'Wooden Wall Clock', 2, 1019, 'TimberTimeCraft', 2100, 6,
NULL);
1 row created.

INSERT INTO Product VALUES (120, 'Rattan Mirror', 2, 1020, 'WovenReflections', 1800, 9, NULL);
1 row created.
```

### Category 3 — Clothing & Accessories

```
INSERT INTO Product VALUES (121, 'Knitted Scarf', 3, 1021, 'CozyYarnCo', 650, 20, NULL);
1 row created.

INSERT INTO Product VALUES (122, 'Crochet Cardigan', 3, 1022, 'StitchAndSoulCraft', 1850, 10,
NULL);
1 row created.

INSERT INTO Product VALUES (123, 'Hand-dyed Tote Bag', 3, 1023, 'IndigoDyeStudio', 720, 15,
NULL);
1 row created.

INSERT INTO Product VALUES (124, 'Embroidered Tunic', 3, 1024, 'NeedleAndThreadCo', 1450, 8,
NULL);
1 row created.

INSERT INTO Product VALUES (125, 'Leather Wallet', 3, 1025, 'HideAndStitch', 980, 18, NULL);
1 row created.

INSERT INTO Product VALUES (126, 'Wool Beanie', 3, 1026, 'WarmThreadsCraft', 420, 30, NULL);
1 row created.

INSERT INTO Product VALUES (127, 'Silk Scarf', 3, 1027, 'SilkRouteWeaves', 890, 14, NULL);
1 row created.

INSERT INTO Product VALUES (128, 'Handmade Sandals', 3, 1028, 'BareStepCraft', 1250, 12, NULL);
1 row created.

INSERT INTO Product VALUES (129, 'Denim Patch Jacket', 3, 1029, 'StitchRebelCo', 2400, 6, NULL);
1 row created.

INSERT INTO Product VALUES (130, 'Beaded Belt', 3, 1030, 'BohoBeltStudio', 560, 20, NULL);
1 row created.
```

#### Category 4 — Art & Collectibles

```
INSERT INTO Product VALUES (131, 'Watercolor Painting', 4, 1031, 'BrushStrokeStudio', 2200, 8, NULL);
1 row created.

INSERT INTO Product VALUES (132, 'Digital Print', 4, 1032, 'PixelPetalArt', 450, 40, NULL);
1 row created.

INSERT INTO Product VALUES (133, 'Hand-carved Sculpture', 4, 1033, 'StoneCraftWorks', 3200, 5, NULL);
1 row created.

INSERT INTO Product VALUES (134, 'Line Art Portrait', 4, 1034, 'InkAndLineStudio', 950, 15, NULL);
1 row created.

INSERT INTO Product VALUES (135, 'Collage Art Piece', 4, 1035, 'MixedMediaCraft', 1650, 9, NULL);
1 row created.

INSERT INTO Product VALUES (136, 'Ceramic Figurine', 4, 1036, 'ClayTalesStudio', 780, 18, NULL);
1 row created.

INSERT INTO Product VALUES (137, 'Framed Photograph', 4, 1037, 'LensAndLightCo', 1100, 12, NULL);
1 row created.

INSERT INTO Product VALUES (138, 'Abstract Canvas Art', 4, 1038, 'ColorFieldStudio', 2600, 7, NULL);
1 row created.

INSERT INTO Product VALUES (139, 'Wooden Carving', 4, 1039, 'TimberTaleCraft', 1350, 10, NULL);
1 row created.

INSERT INTO Product VALUES (140, 'Vintage Postcard Set', 4, 1040, 'PaperTrailArchive', 380, 25, NULL);
1 row created.
```

## Category 5 — Bath & Beauty

```

INSERT INTO Product VALUES (141, 'Handmade Soap Bar', 5, 1041, 'PureLeafSoapCo', 320, 40,
TO_DATE('30-06-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (142, 'Lavender Bath Salts', 5, 1042, 'CalmWatersCraft', 450, 30,
TO_DATE('31-12-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (143, 'Shea Butter Lotion', 5, 1043, 'ButterBloomCo', 550, 25,
TO_DATE('15-08-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (144, 'Beeswax Lip Balm', 5, 1044, 'HoneyHiveCraft', 180, 60,
TO_DATE('30-09-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (145, 'Herbal Face Mask', 5, 1045, 'GreenLeafBeauty', 400, 35,
TO_DATE('31-05-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (146, 'Essential Oil Roller', 5, 1046, 'AromaRootsCraft', 650, 20,
TO_DATE('31-12-2028','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (147, 'Charcoal Soap', 5, 1047, 'PureLeafSoapCo', 350, 40,
TO_DATE('30-06-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (148, 'Body Scrub Jar', 5, 1048, 'SugarAndSaltCo', 480, 28,
TO_DATE('31-10-2027','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (149, 'Solid Shampoo Bar', 5, 1049, 'EcoLatherCraft', 420, 32,
TO_DATE('28-02-2028','DD-MM-YYYY'));
1 row created.

INSERT INTO Product VALUES (150, 'Herbal Bath Tea Bag', 5, 1050, 'SoakAndSteepCo', 300, 45,
TO_DATE('31-07-2027','DD-MM-YYYY'));
1 row created.

```

## Category 6 — Wedding & Party Supplies

```
INSERT INTO Product VALUES (151, 'Wedding Invitation Set', 6, 1051, 'InkAndPaperCo', 850, 20, NULL);
1 row created.

INSERT INTO Product VALUES (152, 'Table Centerpiece', 6, 1052, 'BloomAndBranchStudio', 1450, 10, NULL);
1 row created.

INSERT INTO Product VALUES (153, 'Photo Booth Props', 6, 1053, 'SnapAndSmileCraft', 550, 25, NULL);
1 row created.

INSERT INTO Product VALUES (154, 'Bridal Hair Comb', 6, 1054, 'VeilAndVineStudio', 780, 15, NULL);
1 row created.

INSERT INTO Product VALUES (155, 'Balloon Garland Kit', 6, 1055, 'PartyPopCraft', 620, 20, NULL);
1 row created.

INSERT INTO Product VALUES (156, 'Cake Topper', 6, 1056, 'SweetTopperCo', 350, 30, NULL);
1 row created.

INSERT INTO Product VALUES (157, 'Guest Book', 6, 1057, 'MemoryPagesStudio', 950, 12, NULL);
1 row created.

INSERT INTO Product VALUES (158, 'Ring Bearer Pillow', 6, 1058, 'VelvetVowsCraft', 480, 18, NULL);
1 row created.

INSERT INTO Product VALUES (159, 'Wedding Favor Box Set', 6, 1059, 'FavorAndForm', 600, 25, NULL);
1 row created.

INSERT INTO Product VALUES (160, 'Party Banner Set', 6, 1060, 'StringAndLetterCo', 420, 22, NULL);
1 row created.
```

## Category 7 — Toys & Games

```
INSERT INTO Product VALUES (161, 'Wooden Puzzle', 7, 1061, 'TinkerToyCraft', 650, 20, NULL);
1 row created.

INSERT INTO Product VALUES (162, 'Crochet Stuffed Animal', 7, 1062, 'YarnFriendsCo', 780, 18,
NULL);
1 row created.

INSERT INTO Product VALUES (163, 'Building Block Set', 7, 1063, 'StackAndBuildCraft', 950, 15,
NULL);
1 row created.

INSERT INTO Product VALUES (164, 'Felt Finger Puppets', 7, 1064, 'FeltFableStudio', 380, 30,
NULL);
1 row created.

INSERT INTO Product VALUES (165, 'Wooden Rattle', 7, 1065, 'LittleAcornCraft', 420, 25, NULL);
1 row created.

INSERT INTO Product VALUES (166, 'Cloth Doll', 7, 1066, 'RagAndRibbonCo', 890, 14, NULL);
1 row created.

INSERT INTO Product VALUES (167, 'Memory Match Game', 7, 1067, 'MindfulPlayStudio', 550, 20,
NULL);
1 row created.

INSERT INTO Product VALUES (168, 'Balance Board Toy', 7, 1068, 'WoodenWobbleCraft', 1250, 10,
NULL);
1 row created.

INSERT INTO Product VALUES (169, 'Wooden Xylophone', 7, 1069, 'TinyTuneCraft', 720, 16, NULL);
1 row created.

INSERT INTO Product VALUES (170, 'Knitted Play Blocks', 7, 1070, 'SoftStackCo', 600, 20, NULL);
1 row created.
```

## Category 8 — Craft Supplies & Tools

```
INSERT INTO Product VALUES (171, 'Embroidery Hoop Kit', 8, 1071, 'StitchStartCraft', 450, 30,
NULL);
1 row created.

INSERT INTO Product VALUES (172, 'Yarn Skein Bundle', 8, 1072, 'WoolWindStudio', 380, 40, NULL);
1 row created.

INSERT INTO Product VALUES (173, 'Polymer Clay Set', 8, 1073, 'ClayPlayCo', 550, 25, NULL);
1 row created.

INSERT INTO Product VALUES (174, 'Wood Burning Tool', 8, 1074, 'PyroCraftStudio', 1450, 10,
NULL);
1 row created.

INSERT INTO Product VALUES (175, 'Bead Assortment Box', 8, 1075, 'BeadBoxCraft', 620, 20, NULL);
1 row created.

INSERT INTO Product VALUES (176, 'Leather Crafting Kit', 8, 1076, 'HideWorksStudio', 980, 12,
NULL);
1 row created.

INSERT INTO Product VALUES (177, 'Macrame Cord Set', 8, 1077, 'KnotSupplyCo', 350, 35, NULL);
1 row created.

INSERT INTO Product VALUES (178, 'Fabric Paint Set', 8, 1078, 'ColorThreadCraft', 480, 22,
NULL);
1 row created.

INSERT INTO Product VALUES (179, 'Watercolor Palette', 8, 1079, 'PigmentPathStudio', 720, 18,
NULL);
1 row created.

INSERT INTO Product VALUES (180, 'Wooden Craft Base Set', 8, 1080, 'BaseCraftCo', 550, 20,
NULL);
1 row created.
```

## Category 9 — Vintage Finds

```
INSERT INTO Product VALUES (181, 'Vintage Locket', 9, 1081, 'TimeWornTreasures', 950, 10, NULL);
1 row created.

INSERT INTO Product VALUES (182, 'Antique Brooch', 9, 1082, 'RelicRootsCraft', 780, 12, NULL);
1 row created.

INSERT INTO Product VALUES (183, 'Retro Enamel Pin', 9, 1083, 'PastEraPins', 320, 25, NULL);
1 row created.

INSERT INTO Product VALUES (184, 'Vintage Teacup Set', 9, 1084, 'HeirloomHavenCraft', 1250, 8,
NULL);
1 row created.

INSERT INTO Product VALUES (185, 'Old Map Print', 9, 1085, 'ArchiveAtlasCo', 450, 20, NULL);
1 row created.

INSERT INTO Product VALUES (186, 'Vintage Button Jar', 9, 1086, 'ButtonBoxStudio', 380, 30,
NULL);
1 row created.

INSERT INTO Product VALUES (187, 'Retro Vinyl Record', 9, 1087, 'SpinBackCraft', 650, 15, NULL);
1 row created.

INSERT INTO Product VALUES (188, 'Antique Key Charm', 9, 1088, 'RelicRootsCraft', 300, 28,
NULL);
1 row created.

INSERT INTO Product VALUES (189, 'Vintage Lace Doily', 9, 1089, 'HeirloomHavenCraft', 420, 22,
NULL);
1 row created.

INSERT INTO Product VALUES (190, 'Old Typewriter Keys Set', 9, 1090, 'TypecastTreasures', 780,
12, NULL);
1 row created.
```

## Category 10 — Paper & Party Goods

```
INSERT INTO Product VALUES (191, 'Printable Planner', 10, 1091, 'PagePlanCraft', 250, 50, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (192, 'Greeting Card Set', 10, 1092, 'InkAndPetalStudio', 300, 45,
NULL);
1 row created.
```

```
INSERT INTO Product VALUES (193, 'Origami Paper Pack', 10, 1093, 'FoldCraftCo', 220, 60, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (194, 'Custom Stamp Set', 10, 1094, 'StampAndSealCraft', 450, 30,
NULL);
1 row created.
```

```
INSERT INTO Product VALUES (195, 'Washi Tape Bundle', 10, 1095, 'TapeTrailStudio', 280, 40,
NULL);
1 row created.
```

```
INSERT INTO Product VALUES (196, 'Scrapbook Kit', 10, 1096, 'MemoryLaneCraft', 650, 20, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (197, 'Gift Wrap Set', 10, 1097, 'WrapAndBowCo', 350, 35, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (198, 'Party Invitation Templates', 10, 1098, 'InviteInkStudio', 300,
40, NULL);
1 row created.
```

```
INSERT INTO Product VALUES (199, 'Sticker Sheet Pack', 10, 1099, 'StickerStudioCo', 220, 55,
NULL);
1 row created.
```

```
INSERT INTO Product VALUES (200, 'Kraft Gift Tags', 10, 1100, 'TagAndTwineCraft', 180, 50,
NULL);
1 row created.
```

## 5. Data Retrieval

### 5.1 Display Category Records

```
SELECT * FROM Category;
```

| CATEGORY_ID | CATEGORY_NAME            | DESCRIPTION                         |
|-------------|--------------------------|-------------------------------------|
| 1           | Jewelry                  | Handmade jewelry and accessories    |
| 2           | Home Decor               | Handcrafted home decor items        |
| 3           | Clothing & Accessories   | Handmade clothing and accessories   |
| 4           | Art & Collectibles       | Original art and collectible pieces |
| 5           | Bath & Beauty            | Natural bath and beauty products    |
| 6           | Wedding & Party Supplies | Wedding and party decor supplies    |
| 7           | Toys & Games             | Handmade toys and games             |
| 8           | Craft Supplies & Tools   | Craft supplies and tools            |
| 9           | Vintage Finds            | Vintage and antique items           |
| 10          | Paper & Party Goods      | Paper goods and party printables    |

10 rows selected.

## 5.2 Display Product Records

```
SELECT * FROM Product;
```

### Category 1 — Jewelry

| PROD_ID | PRODUCT_NAME           | CAT_ID | SUPP_ID | BRAND              | PRICE | STOCK | EXPIRY_DATE |
|---------|------------------------|--------|---------|--------------------|-------|-------|-------------|
| 101     | Beaded Necklace        | 1      | 1001    | LunaCraftsCo       | 850   | 25    | -           |
| 102     | Silver Hoop Earrings   | 1      | 1002    | SilverMoonStudio   | 650   | 30    | -           |
| 103     | Charm Bracelet         | 1      | 1003    | WhimsyWorks        | 720   | 20    | -           |
| 104     | Wire Wrapped Ring      | 1      | 1004    | StoneAndWireCo     | 480   | 35    | -           |
| 105     | Pearl Pendant          | 1      | 1005    | OceanPearlCraft    | 950   | 15    | -           |
| 106     | Beaded Anklet          | 1      | 1006    | BohoBeadsShop      | 380   | 40    | -           |
| 107     | Gemstone Choker        | 1      | 1007    | CrystalCoveCraft   | 1100  | 12    | -           |
| 108     | Layered Chain Necklace | 1      | 1008    | GoldenThreadJewels | 890   | 18    | -           |
| 109     | Macrame Bracelet       | 1      | 1009    | KnotsAndBeads      | 320   | 45    | -           |
| 110     | Statement Earrings     | 1      | 1010    | VividVineDesigns   | 560   | 22    | -           |

### Category 2 — Home Decor

| PROD_ID | PRODUCT_NAME         | CAT_ID | SUPP_ID | BRAND              | PRICE | STOCK | EXPIRY_DATE |
|---------|----------------------|--------|---------|--------------------|-------|-------|-------------|
| 111     | Macrame Wall Hanging | 2      | 1011    | KnotsAndBeads      | 1450  | 12    | -           |
| 112     | Ceramic Planter      | 2      | 1012    | ClayAndPetalStudio | 780   | 20    | -           |
| 113     | Wooden Coaster Set   | 2      | 1013    | WoodWhisperCraft   | 550   | 30    | -           |
| 114     | Hand-painted Vase    | 2      | 1014    | BrushAndClayArt    | 1650  | 8     | -           |
| 115     | Woven Basket         | 2      | 1015    | WillowWeaveCo      | 920   | 15    | -           |
| 116     | Candle Holder        | 2      | 1016    | FlameAndForm       | 600   | 25    | -           |
| 117     | Dreamcatcher         | 2      | 1017    | FeatherDreamsCraft | 480   | 28    | -           |
| 118     | Wall Tapestry        | 2      | 1018    | TextileTalesShop   | 1350  | 10    | -           |
| 119     | Wooden Wall Clock    | 2      | 1019    | TimberTimeCraft    | 2100  | 6     | -           |
| 120     | Rattan Mirror        | 2      | 1020    | WovenReflections   | 1800  | 9     | -           |

### Category 3 — Clothing & Accessories

| PROD_ID | PRODUCT_NAME       | CAT_ID | SUPP_ID | BRAND              | PRICE | STOCK | EXPIRY_DATE |
|---------|--------------------|--------|---------|--------------------|-------|-------|-------------|
| 121     | Knitted Scarf      | 3      | 1021    | CozyYarnCo         | 650   | 20    | -           |
| 122     | Crochet Cardigan   | 3      | 1022    | StitchAndSoulCraft | 1850  | 10    | -           |
| 123     | Hand-dyed Tote Bag | 3      | 1023    | IndigoDyeStudio    | 720   | 15    | -           |
| 124     | Embroidered Tunic  | 3      | 1024    | NeedleAndThreadCo  | 1450  | 8     | -           |
| 125     | Leather Wallet     | 3      | 1025    | HideAndStitch      | 980   | 18    | -           |
| 126     | Wool Beanie        | 3      | 1026    | WarmThreadsCraft   | 420   | 30    | -           |
| 127     | Silk Scarf         | 3      | 1027    | SilkRouteWeaves    | 890   | 14    | -           |
| 128     | Handmade Sandals   | 3      | 1028    | BareStepCraft      | 1250  | 12    | -           |
| 129     | Denim Patch Jacket | 3      | 1029    | StitchRebelCo      | 2400  | 6     | -           |
| 130     | Beaded Belt        | 3      | 1030    | BohoBeltStudio     | 560   | 20    | -           |

### Category 4 — Art & Collectibles

| PROD_ID | PRODUCT_NAME          | CAT_ID | SUPP_ID | BRAND             | PRICE | STOCK | EXPIRY_DATE |
|---------|-----------------------|--------|---------|-------------------|-------|-------|-------------|
| 131     | Watercolor Painting   | 4      | 1031    | BrushStrokeStudio | 2200  | 8     | -           |
| 132     | Digital Print         | 4      | 1032    | PixelPetalArt     | 450   | 40    | -           |
| 133     | Hand-carved Sculpture | 4      | 1033    | StoneCraftWorks   | 3200  | 5     | -           |
| 134     | Line Art Portrait     | 4      | 1034    | InkAndLineStudio  | 950   | 15    | -           |
| 135     | Collage Art Piece     | 4      | 1035    | MixedMediaCraft   | 1650  | 9     | -           |
| 136     | Ceramic Figurine      | 4      | 1036    | ClayTalesStudio   | 780   | 18    | -           |
| 137     | Framed Photograph     | 4      | 1037    | LensAndLightCo    | 1100  | 12    | -           |
| 138     | Abstract Canvas Art   | 4      | 1038    | ColorFieldStudio  | 2600  | 7     | -           |
| 139     | Wooden Carving        | 4      | 1039    | TimberTaleCraft   | 1350  | 10    | -           |
| 140     | Vintage Postcard Set  | 4      | 1040    | PaperTrailArchive | 380   | 25    | -           |

### Category 5 — Bath & Beauty

| PROD_ID | PRODUCT_NAME      | CAT_ID | SUPP_ID | BRAND          | PRICE | STOCK | EXPIRY_DATE |
|---------|-------------------|--------|---------|----------------|-------|-------|-------------|
| 141     | Handmade Soap Bar | 5      | 1041    | PureLeafSoapCo | 320   | 40    | 30-JUN-27   |

| PROD_ID | PRODUCT_NAME         | CAT_ID | SUPP_ID | BRAND           | PRICE | STOCK | EXPIRY_DATE |
|---------|----------------------|--------|---------|-----------------|-------|-------|-------------|
| 142     | Lavender Bath Salts  | 5      | 1042    | CalmWatersCraft | 450   | 30    | 31-DEC-27   |
| 143     | Shea Butter Lotion   | 5      | 1043    | ButterBloomCo   | 550   | 25    | 15-AUG-27   |
| 144     | Beeswax Lip Balm     | 5      | 1044    | HoneyHiveCraft  | 180   | 60    | 30-SEP-27   |
| 145     | Herbal Face Mask     | 5      | 1045    | GreenLeafBeauty | 400   | 35    | 31-MAY-27   |
| 146     | Essential Oil Roller | 5      | 1046    | AromaRootsCraft | 650   | 20    | 31-DEC-28   |
| 147     | Charcoal Soap        | 5      | 1047    | PureLeafSoapCo  | 350   | 40    | 30-JUN-27   |
| 148     | Body Scrub Jar       | 5      | 1048    | SugarAndSaltCo  | 480   | 28    | 31-OCT-27   |
| 149     | Solid Shampoo Bar    | 5      | 1049    | EcoLatherCraft  | 420   | 32    | 28-FEB-28   |
| 150     | Herbal Bath Tea Bag  | 5      | 1050    | SoakAndSteepCo  | 300   | 45    | 31-JUL-27   |

### Category 6 — Wedding & Party Supplies

| PROD_ID | PRODUCT_NAME           | CAT_ID | SUPP_ID | BRAND                | PRICE | STOCK | EXPIRY_DATE |
|---------|------------------------|--------|---------|----------------------|-------|-------|-------------|
| 151     | Wedding Invitation Set | 6      | 1051    | InkAndPaperCo        | 850   | 20    | -           |
| 152     | Table Centerpiece      | 6      | 1052    | BloomAndBranchStudio | 1450  | 10    | -           |
| 153     | Photo Booth Props      | 6      | 1053    | SnapAndSmileCraft    | 550   | 25    | -           |
| 154     | Bridal Hair Comb       | 6      | 1054    | VeilAndVineStudio    | 780   | 15    | -           |
| 155     | Balloon Garland Kit    | 6      | 1055    | PartyPopCraft        | 620   | 20    | -           |
| 156     | Cake Topper            | 6      | 1056    | SweetTopperCo        | 350   | 30    | -           |
| 157     | Guest Book             | 6      | 1057    | MemoryPagesStudio    | 950   | 12    | -           |
| 158     | Ring Bearer Pillow     | 6      | 1058    | VelvetVowsCraft      | 480   | 18    | -           |
| 159     | Wedding Favor Box Set  | 6      | 1059    | FavorAndForm         | 600   | 25    | -           |
| 160     | Party Banner Set       | 6      | 1060    | StringAndLetterCo    | 420   | 22    | -           |

### Category 7 — Toys & Games

| PROD_ID | PRODUCT_NAME           | CAT_ID | SUPP_ID | BRAND              | PRICE | STOCK | EXPIRY_DATE |
|---------|------------------------|--------|---------|--------------------|-------|-------|-------------|
| 161     | Wooden Puzzle          | 7      | 1061    | TinkerToyCraft     | 650   | 20    | -           |
| 162     | Crochet Stuffed Animal | 7      | 1062    | YarnFriendsCo      | 780   | 18    | -           |
| 163     | Building Block Set     | 7      | 1063    | StackAndBuildCraft | 950   | 15    | -           |
| 164     | Felt Finger Puppets    | 7      | 1064    | FeltFableStudio    | 380   | 30    | -           |
| 165     | Wooden Rattle          | 7      | 1065    | LittleAcornCraft   | 420   | 25    | -           |
| 166     | Cloth Doll             | 7      | 1066    | RagAndRibbonCo     | 890   | 14    | -           |
| 167     | Memory Match Game      | 7      | 1067    | MindfulPlayStudio  | 550   | 20    | -           |
| 168     | Balance Board Toy      | 7      | 1068    | WoodenWobbleCraft  | 1250  | 10    | -           |
| 169     | Wooden Xylophone       | 7      | 1069    | TinyTuneCraft      | 720   | 16    | -           |
| 170     | Knitted Play Blocks    | 7      | 1070    | SoftStackCo        | 600   | 20    | -           |

### Category 8 — Craft Supplies & Tools

| PROD_ID | PRODUCT_NAME          | CAT_ID | SUPP_ID | BRAND             | PRICE | STOCK | EXPIRY_DATE |
|---------|-----------------------|--------|---------|-------------------|-------|-------|-------------|
| 171     | Embroidery Hoop Kit   | 8      | 1071    | StitchStartCraft  | 450   | 30    | -           |
| 172     | Yarn Skein Bundle     | 8      | 1072    | WoolWindStudio    | 380   | 40    | -           |
| 173     | Polymer Clay Set      | 8      | 1073    | ClayPlayCo        | 550   | 25    | -           |
| 174     | Wood Burning Tool     | 8      | 1074    | PyroCraftStudio   | 1450  | 10    | -           |
| 175     | Bead Assortment Box   | 8      | 1075    | BeadBoxCraft      | 620   | 20    | -           |
| 176     | Leather Crafting Kit  | 8      | 1076    | HideWorksStudio   | 980   | 12    | -           |
| 177     | Macrame Cord Set      | 8      | 1077    | KnotSupplyCo      | 350   | 35    | -           |
| 178     | Fabric Paint Set      | 8      | 1078    | ColorThreadCraft  | 480   | 22    | -           |
| 179     | Watercolor Palette    | 8      | 1079    | PigmentPathStudio | 720   | 18    | -           |
| 180     | Wooden Craft Base Set | 8      | 1080    | BaseCraftCo       | 550   | 20    | -           |

### Category 9 — Vintage Finds

| PROD_ID | PRODUCT_NAME            | CAT_ID | SUPP_ID | BRAND              | PRICE | STOCK | EXPIRY_DATE |
|---------|-------------------------|--------|---------|--------------------|-------|-------|-------------|
| 181     | Vintage Locket          | 9      | 1081    | TimeWornTreasures  | 950   | 10    | -           |
| 182     | Antique Brooch          | 9      | 1082    | RelicRootsCraft    | 780   | 12    | -           |
| 183     | Retro Enamel Pin        | 9      | 1083    | PastEraPins        | 320   | 25    | -           |
| 184     | Vintage Teacup Set      | 9      | 1084    | HeirloomHavenCraft | 1250  | 8     | -           |
| 185     | Old Map Print           | 9      | 1085    | ArchiveAtlasCo     | 450   | 20    | -           |
| 186     | Vintage Button Jar      | 9      | 1086    | ButtonBoxStudio    | 380   | 30    | -           |
| 187     | Retro Vinyl Record      | 9      | 1087    | SpinBackCraft      | 650   | 15    | -           |
| 188     | Antique Key Charm       | 9      | 1088    | RelicRootsCraft    | 300   | 28    | -           |
| 189     | Vintage Lace Doily      | 9      | 1089    | HeirloomHavenCraft | 420   | 22    | -           |
| 190     | Old Typewriter Keys Set | 9      | 1090    | TypecastTreasures  | 780   | 12    | -           |

## Category 10 — Paper & Party Goods

| PROD_ID | PRODUCT_NAME               | CAT_ID | SUPP_ID | BRAND             | PRICE | STOCK | EXPIRY_DATE |
|---------|----------------------------|--------|---------|-------------------|-------|-------|-------------|
| 191     | Printable Planner          | 10     | 1091    | PagePlanCraft     | 250   | 50    | -           |
| 192     | Greeting Card Set          | 10     | 1092    | InkAndPetalStudio | 300   | 45    | -           |
| 193     | Origami Paper Pack         | 10     | 1093    | FoldCraftCo       | 220   | 60    | -           |
| 194     | Custom Stamp Set           | 10     | 1094    | StampAndSealCraft | 450   | 30    | -           |
| 195     | Washi Tape Bundle          | 10     | 1095    | TapeTrailStudio   | 280   | 40    | -           |
| 196     | Scrapbook Kit              | 10     | 1096    | MemoryLaneCraft   | 650   | 20    | -           |
| 197     | Gift Wrap Set              | 10     | 1097    | WrapAndBowCo      | 350   | 35    | -           |
| 198     | Party Invitation Templates | 10     | 1098    | InviteInkStudio   | 300   | 40    | -           |
| 199     | Sticker Sheet Pack         | 10     | 1099    | StickerStudioCo   | 220   | 55    | -           |
| 200     | Kraft Gift Tags            | 10     | 1100    | TagAndTwineCraft  | 180   | 50    | -           |

100 rows selected.

## 6. Product Update

Update the price and stock of the Knitted Scarf (Product\_ID 121), the Digital Print (Product\_ID 132) and the Party Banner Set (Product\_ID 160).

```
UPDATE Product
SET Price = 700, Stock = 25
WHERE Product_ID = 121;
1 row updated.

UPDATE Product
SET Price = 500,
    Stock = 35
WHERE Product_ID = 132;
1 row updated.

UPDATE Product
SET Price = 450,
    Stock = 28
WHERE Product_ID = 160;
1 row updated.

SELECT *
FROM Product
WHERE Product_ID IN (121, 132, 160);
```

| PROD_ID | PRODUCT_NAME     | CAT_ID | SUPP_ID | BRAND             | PRICE | STOCK | EXPIRY_DATE |
|---------|------------------|--------|---------|-------------------|-------|-------|-------------|
| 121     | Knitted Scarf    | 3      | 1021    | CozyYarnCo        | 700   | 25    | -           |
| 132     | Digital Print    | 4      | 1032    | PixelPetalArt     | 500   | 35    | -           |
| 160     | Party Banner Set | 6      | 1060    | StringAndLetterCo | 450   | 28    | -           |

*Verify the updated price and stock by selecting the records.*

## 7. Product Deletion

Delete the product Handmade Sandals (Product\_ID 118), Yarn Skein Bundle (Product\_ID 172) and Greeting Card Set (Product\_ID 192) from the Product table.

```
DELETE FROM Product
WHERE Product_ID = 118;
1 row deleted.
```

```
DELETE FROM Product
WHERE Product_ID = 172;
1 row deleted.
```

```
DELETE FROM Product
WHERE Product_ID = 192;
1 row deleted.
```

Verify the deletion by displaying the remaining product records.

```
SELECT * FROM Product;

97 rows selected.
```

The output matches the table in Section 5.2 with the three deleted rows (118, 172, 192) removed.

## 8. Category-wise Product Reports

### 8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category, using an inner JOIN between Category and Product.

```
SELECT c.Category_ID, c.Category_Name, p.Product_ID, p.Product_Name, p.Price, p.Stock
FROM Category c JOIN Product p
    ON c.Category_ID = p.Category_ID
ORDER BY c.Category_ID, p.Product_ID;
```

#### 1 — Jewelry

| PRODUCT_ID | PRODUCT_NAME           | PRICE | STOCK |
|------------|------------------------|-------|-------|
| 101        | Beaded Necklace        | 850   | 25    |
| 102        | Silver Hoop Earrings   | 650   | 30    |
| 103        | Charm Bracelet         | 720   | 20    |
| 104        | Wire Wrapped Ring      | 480   | 35    |
| 105        | Pearl Pendant          | 950   | 15    |
| 106        | Beaded Anklet          | 380   | 40    |
| 107        | Gemstone Choker        | 1100  | 12    |
| 108        | Layered Chain Necklace | 890   | 18    |
| 109        | Macrame Bracelet       | 320   | 45    |
| 110        | Statement Earrings     | 560   | 22    |

#### 2 — Home Decor

| PRODUCT_ID | PRODUCT_NAME         | PRICE | STOCK |
|------------|----------------------|-------|-------|
| 111        | Macrame Wall Hanging | 1450  | 12    |
| 112        | Ceramic Planter      | 780   | 20    |
| 113        | Wooden Coaster Set   | 550   | 30    |
| 114        | Hand-painted Vase    | 1650  | 8     |
| 115        | Woven Basket         | 920   | 15    |
| 116        | Candle Holder        | 600   | 25    |
| 117        | Dreamcatcher         | 480   | 28    |
| 118        | Wall Tapestry        | 1350  | 10    |
| 119        | Wooden Wall Clock    | 2100  | 6     |
| 120        | Rattan Mirror        | 1800  | 9     |

### 3 — Clothing & Accessories

| PRODUCT_ID | PRODUCT_NAME       | PRICE | STOCK |
|------------|--------------------|-------|-------|
| 121        | Knitted Scarf      | 650   | 20    |
| 122        | Crochet Cardigan   | 1850  | 10    |
| 123        | Hand-dyed Tote Bag | 720   | 15    |
| 124        | Embroidered Tunic  | 1450  | 8     |
| 125        | Leather Wallet     | 980   | 18    |
| 126        | Wool Beanie        | 420   | 30    |
| 127        | Silk Scarf         | 890   | 14    |
| 128        | Handmade Sandals   | 1250  | 12    |
| 129        | Denim Patch Jacket | 2400  | 6     |
| 130        | Beaded Belt        | 560   | 20    |

### 4 — Art & Collectibles

| PRODUCT_ID | PRODUCT_NAME          | PRICE | STOCK |
|------------|-----------------------|-------|-------|
| 131        | Watercolor Painting   | 2200  | 8     |
| 132        | Digital Print         | 450   | 40    |
| 133        | Hand-carved Sculpture | 3200  | 5     |
| 134        | Line Art Portrait     | 950   | 15    |
| 135        | Collage Art Piece     | 1650  | 9     |
| 136        | Ceramic Figurine      | 780   | 18    |
| 137        | Framed Photograph     | 1100  | 12    |
| 138        | Abstract Canvas Art   | 2600  | 7     |
| 139        | Wooden Carving        | 1350  | 10    |
| 140        | Vintage Postcard Set  | 380   | 25    |

### 5 — Bath & Beauty

| PRODUCT_ID | PRODUCT_NAME         | PRICE | STOCK |
|------------|----------------------|-------|-------|
| 141        | Handmade Soap Bar    | 320   | 40    |
| 142        | Lavender Bath Salts  | 450   | 30    |
| 143        | Shea Butter Lotion   | 550   | 25    |
| 144        | Beeswax Lip Balm     | 180   | 60    |
| 145        | Herbal Face Mask     | 400   | 35    |
| 146        | Essential Oil Roller | 650   | 20    |
| 147        | Charcoal Soap        | 350   | 40    |
| 148        | Body Scrub Jar       | 480   | 28    |
| 149        | Solid Shampoo Bar    | 420   | 32    |
| 150        | Herbal Bath Tea Bag  | 300   | 45    |

### 6 — Wedding & Party Supplies

| PRODUCT_ID | PRODUCT_NAME           | PRICE | STOCK |
|------------|------------------------|-------|-------|
| 151        | Wedding Invitation Set | 850   | 20    |
| 152        | Table Centerpiece      | 1450  | 10    |
| 153        | Photo Booth Props      | 550   | 25    |
| 154        | Bridal Hair Comb       | 780   | 15    |
| 155        | Balloon Garland Kit    | 620   | 20    |
| 156        | Cake Topper            | 350   | 30    |
| 157        | Guest Book             | 950   | 12    |
| 158        | Ring Bearer Pillow     | 480   | 18    |
| 159        | Wedding Favor Box Set  | 600   | 25    |
| 160        | Party Banner Set       | 420   | 22    |

### 7 — Toys & Games

| PRODUCT_ID | PRODUCT_NAME           | PRICE | STOCK |
|------------|------------------------|-------|-------|
| 161        | Wooden Puzzle          | 650   | 20    |
| 162        | Crochet Stuffed Animal | 780   | 18    |
| 163        | Building Block Set     | 950   | 15    |
| 164        | Felt Finger Puppets    | 380   | 30    |
| 165        | Wooden Rattle          | 420   | 25    |
| 166        | Cloth Doll             | 890   | 14    |

| PRODUCT_ID | PRODUCT_NAME        | PRICE | STOCK |
|------------|---------------------|-------|-------|
| 167        | Memory Match Game   | 550   | 20    |
| 168        | Balance Board Toy   | 1250  | 10    |
| 169        | Wooden Xylophone    | 720   | 16    |
| 170        | Knitted Play Blocks | 600   | 20    |

### 8 — Craft Supplies & Tools

| PRODUCT_ID | PRODUCT_NAME          | PRICE | STOCK |
|------------|-----------------------|-------|-------|
| 171        | Embroidery Hoop Kit   | 450   | 30    |
| 172        | Yarn Skein Bundle     | 380   | 40    |
| 173        | Polymer Clay Set      | 550   | 25    |
| 174        | Wood Burning Tool     | 1450  | 10    |
| 175        | Bead Assortment Box   | 620   | 20    |
| 176        | Leather Crafting Kit  | 980   | 12    |
| 177        | Macrame Cord Set      | 350   | 35    |
| 178        | Fabric Paint Set      | 480   | 22    |
| 179        | Watercolor Palette    | 720   | 18    |
| 180        | Wooden Craft Base Set | 550   | 20    |

### 9 — Vintage Finds

| PRODUCT_ID | PRODUCT_NAME            | PRICE | STOCK |
|------------|-------------------------|-------|-------|
| 181        | Vintage Locket          | 950   | 10    |
| 182        | Antique Brooch          | 780   | 12    |
| 183        | Retro Enamel Pin        | 320   | 25    |
| 184        | Vintage Teacup Set      | 1250  | 8     |
| 185        | Old Map Print           | 450   | 20    |
| 186        | Vintage Button Jar      | 380   | 30    |
| 187        | Retro Vinyl Record      | 650   | 15    |
| 188        | Antique Key Charm       | 300   | 28    |
| 189        | Vintage Lace Doily      | 420   | 22    |
| 190        | Old Typewriter Keys Set | 780   | 12    |

### 10 — Paper & Party Goods

| PRODUCT_ID | PRODUCT_NAME               | PRICE | STOCK |
|------------|----------------------------|-------|-------|
| 191        | Printable Planner          | 250   | 50    |
| 192        | Greeting Card Set          | 300   | 45    |
| 193        | Origami Paper Pack         | 220   | 60    |
| 194        | Custom Stamp Set           | 450   | 30    |
| 195        | Washi Tape Bundle          | 280   | 40    |
| 196        | Scrapbook Kit              | 650   | 20    |
| 197        | Gift Wrap Set              | 350   | 35    |
| 198        | Party Invitation Templates | 300   | 40    |
| 199        | Sticker Sheet Pack         | 220   | 55    |
| 200        | Kraft Gift Tags            | 180   | 50    |

97 rows selected.

### Conclusion

The Category and Product tables were successfully designed, created, and populated with sample data. CRUD operations (Insert, Select, Update, Delete) were performed and verified, and a JOIN query was used to generate a category-wise product report, demonstrating a working one-to-many relationship between Category and Product.